

HW4

November 2, 2025

1 Homework 4

In this homework, we will cover panorama stitching.

This homework was adapted from Stanford CS131.

1.1 Setup

```
[ ]: import os

if not os.path.exists("CS485_584"):
    # Clone the repository if it doesn't already exist
    !git clone https://github.com/YoungjoongEmory/CS485_584.git

%cd CS485_584/Assignments/HW4/
```

```
Cloning into 'CS485_584'...
remote: Enumerating objects: 174, done.
remote: Counting objects: 100% (174/174), done.
remote: Compressing objects: 100% (161/161), done.
remote: Total 174 (delta 44), reused 7 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (174/174), 13.53 MiB | 12.19 MiB/s, done.
Resolving deltas: 100% (44/44), done.
/content/CS485_584/Assignments/HW4
```

```
[ ]: # Install the necessary dependencies
# (restart your runtime session if prompted to, and then re-run this cell)
!pip install -r requirements.txt
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from -r requirements.txt (line 1)) (2.0.2)
Requirement already satisfied: scikit-image in /usr/local/lib/python3.12/dist-
packages (from -r requirements.txt (line 2)) (0.25.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from -r requirements.txt (line 3)) (1.16.3)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from -r requirements.txt (line 4)) (3.10.0)
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-
packages (from -r requirements.txt (line 5)) (4.12.0.88)
```

Requirement already satisfied: networkx>=3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-image->-r requirements.txt (line 2)) (3.5)

Requirement already satisfied: pillow>=10.1 in /usr/local/lib/python3.12/dist-packages (from scikit-image->-r requirements.txt (line 2)) (11.3.0)

Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/python3.12/dist-packages (from scikit-image->-r requirements.txt (line 2)) (2.37.0)

Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib/python3.12/dist-packages (from scikit-image->-r requirements.txt (line 2)) (2025.10.16)

Requirement already satisfied: packaging>=21 in /usr/local/lib/python3.12/dist-packages (from scikit-image->-r requirements.txt (line 2)) (25.0)

Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image->-r requirements.txt (line 2)) (0.4)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r requirements.txt (line 4)) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r requirements.txt (line 4)) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r requirements.txt (line 4)) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r requirements.txt (line 4)) (1.4.9)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r requirements.txt (line 4)) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r requirements.txt (line 4)) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->-r requirements.txt (line 4)) (1.17.0)

```
[ ]: from __future__ import print_function

import numpy as np
from skimage import filters
from skimage.feature import corner_peaks
from skimage.io import imread
from scipy.spatial.distance import cdist
from scipy.ndimage import convolve
from scipy import ndimage
import matplotlib.pyplot as plt
import cv2 as cv
```

```

from utils import pad, unpad, get_output_space, warp_image, plot_matches,
↳ describe_keypoints

%matplotlib inline
plt.rcParams['figure.figsize'] = (12.0, 9.0)
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

```

2 Panorama Stitching

Panorama stitching is an early success of computer vision. Matthew Brown and David G. Lowe published a famous [panoramic image stitching paper](#) in 2007. Since then, automatic panorama stitching technology has been widely adopted in many applications such as Google Street View, panorama photos on smartphones, and stitching software such as Photosynth and AutoStitch.

We will detect and match keypoints from multiple images to build a single panoramic image. This will involve several tasks: 1. Compare two sets of descriptors coming from two different images and find matching keypoints. 2. Given matching keypoints, use least-squares to find the affine transformation matrix that maps points in one image to another. 3. Use RANSAC to give a more robust estimate of the affine transformation matrix. Given the transformation matrix, use it to transform the second image and overlay it on the first image, forming a panorama. 4. Blend panorama images together to remove blurry regions of overlapping images.

2.1 1. Describing and Matching Keypoints (0 points)

We will use Harris corner detection to detect keypoints in two images. Given these two sets of keypoints, we will then determine which pairs of keypoints come from the same 3D points projected onto the two images. We do so by first converting the region around each keypoint into a descriptor. Then, we find good matches in the two sets of descriptors based on Euclidean distance.

Use the following definition of cornerness response:

$$\theta = \det(M) - \alpha(\text{trace}(M))^2 = (w * I_x^2)(w * I_y^2) - (w * (I_x I_y))^2 - \alpha(w * (I_x^2) + w * (I_y^2))^2$$

where (*) denotes convolution and (w) is the window function.

```

[17]: def harris_corners(img, window_size=3, k=0.04):
      """
      Compute Harris corner response map. Follow the math equation
      R=Det(M)-k(Trace(M)^2).
      Hint:
      You may use the function ndimage.convolve,
      which is already imported above. If you use convolve(), remember to
      specify zero-padding to match our equations, for example:

      out_image = ndimage.convolve(in_image, kernel, mode='constant',
      ↳ cval=0)

```

```

Args:
    img: Grayscale image of shape (H, W)
    window_size: size of the window function
    k: sensitivity parameter

Returns:
    response: Harris response image of shape (H, W)
    """

H, W = img.shape
window = np.ones((window_size, window_size)) # window function

response = np.zeros((H, W))

# 1. Compute x and y derivatives (I_x, I_y) of an image
dx = filters.sobel_v(img)
dy = filters.sobel_h(img)

### YOUR CODE HERE

# Compute products of derivatives using dx and dy.
Ixx = dx * dx
Iyy = dy * dy
Ixy = dx * dy

# Sum of products within the window (use zero padding)
Sxx = ndimage.convolve(Ixx, window, mode='constant', cval=0)
Syy = ndimage.convolve(Iyy, window, mode='constant', cval=0)
Sxy = ndimage.convolve(Ixy, window, mode='constant', cval=0)

# Harris response:  $R = \det(M) - k * (\text{trace}(M))^2$ 
detM = Sxx * Syy - Sxy ** 2
traceM = Sxx + Syy
response = detM - k * (traceM ** 2)

### END YOUR CODE

return response

def simple_descriptor(patch):
    """
    Describe the patch by normalizing the image values into a standard
    normal distribution (having mean of 0 and standard deviation of 1)
    and then flattening into a 1D array.

    The normalization will make the descriptor more robust to change

```

in lighting condition.

Hint:

In this case of normalization, if a denominator is zero, divide by 1_
↳ instead.

Args:

patch: grayscale image patch of shape (H, W)

Returns:

*feature: 1D array of shape (H * W)*

"""

```
feature = []  
patch = patch.astype(np.float32)  
mu = np.mean(patch)  
sigma = np.std(patch)
```

YOUR CODE HERE

```
if sigma == 0:  
    sigma = 1
```

```
norm = (patch - mu) / (sigma)
```

END YOUR CODE

```
feature = norm.flatten()
```

```
return feature
```

```
def match_descriptors(desc1, desc2, threshold=0.5):
```

"""

Match the feature descriptors by finding distances between them. A match is_
↳ formed

when the distance to the closest vector is much smaller than the distance_
↳ to the

second-closest, that is, the ratio of the distances should be STRICTLY_
↳ SMALLER

than the threshold (NOT equal to). Return the matches as pairs of vector_
↳ indices.

Hint:

The Numpy functions np.sort, np.argmin, np.asarray might be useful

*The Scipy function cdist calculates Euclidean distance between all
pairs of inputs*

Args:

```

    desc1: an array of shape (M, P) holding descriptors of size P about M_
↳keypoints
    desc2: an array of shape (N, P) holding descriptors of size P about N_
↳keypoints

    Returns:
        matches: an array of shape (Q, 2) where each row holds the indices of_
↳one pair
        of matching descriptors
    """

    matches = []

    M = desc1.shape[0]

    # dists is (M, N); each row corresponds to one descriptor from image 1
    # and distances to all descriptors in image 2
    dists = cdist(desc1, desc2)

    # For each descriptor in desc1, apply ratio test against desc2

    # dists is (M, N); for each row, find indices of two smallest distances
    idx_sorted = np.argsort(dists, axis=1)
    nearest = idx_sorted[:, 0]
    second_nearest = idx_sorted[:, 1]

    # Gather corresponding smallest and second smallest distances
    d1 = dists[np.arange(M), nearest]
    d2 = dists[np.arange(M), second_nearest]

    # Ratio test
    valid = d1 / (d2 + 1e-12) < threshold
    for i in np.where(valid)[0]:
        matches.append([i, nearest[i]])

    matches = np.asarray(matches, dtype=int) if len(matches) > 0 else np.
↳zeros((0, 2), dtype=int)

    return matches

```

Run the following cell to detect and match the keypoints in two images.

```

[18]: img1 = imread('uttower1.jpg', as_gray=True)
      img2 = imread('uttower2.jpg', as_gray=True)

      np.random.seed(131)

```

```

# Detect keypoints in two images
keypoints1 = corner_peaks(harris_corners(img1, window_size=3),
                           threshold_rel=0.05,
                           exclude_border=8)
keypoints2 = corner_peaks(harris_corners(img2, window_size=3),
                           threshold_rel=0.05,
                           exclude_border=8)

# Extract features from the corners
desc1 = describe_keypoints(img1, keypoints1,
                            desc_func=simple_descriptor,
                            patch_size=5)
desc2 = describe_keypoints(img2, keypoints2,
                            desc_func=simple_descriptor,
                            patch_size=5)

# Match descriptors in image1 to those in image2
matches = match_descriptors(desc1, desc2, 0.7)

```

2.2 2. Transformation Estimation (20 points)

Now, we will use these matched keypoints to find a **transformation matrix** that maps points in the second image to the corresponding coordinates in the first image. In other words, if the point $p_1 = [y_1, x_1]$ in image 1 matches with $p_2 = [y_2, x_2]$ in image 2, we need to find an affine transformation matrix H such that

$$\tilde{p}_2 H = \tilde{p}_1,$$

where $\tilde{p}_1$ and $\tilde{p}_2$ are homogenous coordinates of p_1 and p_2 .

Note that it may be impossible to find the transformation H that maps every point in image 2 exactly to the corresponding point in image 1. However, **we can estimate the transformation matrix with the least squares method**. Given N matched keypoint pairs, let X_1 and X_2 be $N \times 3$ matrices whose rows are homogenous coordinates of corresponding keypoints in image 1 and image 2 respectively. Then, we can estimate H by solving the least squares problem,

$$X_2 H = X_1$$

Implement `fit_affine_matrix` below.

Hint: read the [documentation](#) for `np.linalg.lstsq`

```

[19]: def fit_affine_matrix(p1, p2, to_pad=True):
      """
      Fit affine matrix such that  $p2 * H = p1$ . First, pad the descriptor vectors
      with a 1 using pad() to convert to homogeneous coordinates, then return
      the least squares fit affine matrix in homogeneous coordinates.

```

Hint:

You can use `np.linalg.lstsq` function to solve the problem.

Explicitly specify `np.linalg.lstsq`'s new default parameter `rcond=None` to suppress deprecation warnings.

Args:

*p1: an array of shape (M, P) holding descriptors of size P about $M_{\square}$
↪keypoints*

*p2: an array of shape (M, P) holding descriptors of size P about $M_{\square}$
↪keypoints*

Return:

H: a matrix of shape $(P+1, P+1)$ that transforms $p2$ to $p1$ in homogeneous coordinates

"""

```
assert (p1.shape[0] == p2.shape[0]),\  
    'Different number of points in p1 and p2'
```

```
if to_pad:  
    p1 = pad(p1)  
    p2 = pad(p2)
```

```
### YOUR CODE HERE
```

```
H = np.linalg.lstsq(p2, p1, rcond=None)[0]
```

```
### END YOUR CODE
```

```
# Sometimes numerical issues cause least-squares to produce the last  
# column which is not exactly [0, 0, 1]
```

```
H[:,2] = np.array([0, 0, 1])
```

```
return H
```

[20]: *# Sanity check for `fit_affine_matrix`*

```
# Test inputs
```

```
a = np.array([[0.5, 0.1], [0.4, 0.2], [0.8, 0.2]])
```

```
b = np.array([[0.3, -0.2], [-0.4, -0.9], [0.1, 0.1]])
```

```
H = fit_affine_matrix(b, a)
```

```
# Target output
```

```
sol = np.array(  
    [[1.25, 2.5, 0.0],  
     [-5.75, -4.5, 0.0],
```

```

        [0.25, -1.0, 1.0]]
    )

    error = np.sum((H - sol) ** 2)

    if error < 1e-20:
        print('Implementation correct!')
    else:
        print('There is something wrong.')

```

Implementation correct!

After checking that your `fit_affine_matrix` function is running correctly, run the following code to apply it to images.

Images will be warped and image 2 will be mapped to image 1.

```

[21]: p1 = keypoints1[matches[:,0]]
      p2 = keypoints2[matches[:,1]]

      # Find affine transformation matrix H that maps p2 to p1
      H = fit_affine_matrix(p1, p2)

      output_shape, offset = get_output_space(img1, [img2], [H])
      print("Output shape:", output_shape)
      print("Offset:", offset)

      # Warp images into output space
      img1_warped = warp_image(img1, np.eye(3), output_shape, offset)
      img1_mask = (img1_warped != -1) # Mask == 1 inside the image
      img1_warped[~img1_mask] = 0      # Return background values to 0

      img2_warped = warp_image(img2, H, output_shape, offset)
      img2_mask = (img2_warped != -1) # Mask == 1 inside the image
      img2_warped[~img2_mask] = 0      # Return background values to 0

      # Plot warped images
      plt.subplot(1,2,1)
      plt.imshow(img1_warped)
      plt.title('Image 1 Warped')
      plt.axis('off')

      plt.subplot(1,2,2)
      plt.imshow(img2_warped)
      plt.title('Image 2 Warped')
      plt.axis('off')

      plt.show()

```

Output shape: [493 631]
Offset: [-37.17230306 0.]



Next, the two warped images are merged to get a panorama.

Your panorama may not look good at this point, but we will later use other techniques to get a better result!

```
[22]: # Merge the two images
merged = img1_warped + img2_warped

# Track the overlap by adding the masks together
overlap = (img1_mask * 1.0 + # Multiply by 1.0 for bool -> float conversion
           img2_mask)

# Normalize through division by `overlap` - but ensure the minimum is 1
normalized = merged / np.maximum(overlap, 1)

plt.imshow(normalized)
plt.axis('off')
plt.title('Fit-Affine Panorama')
plt.show()

plt.imshow(imread('solution_fit_affine_panorama.png'))
plt.axis('off')
plt.title('Fit-Affine Panorama Solution')
plt.show()
```

Fit-Affine Panorama





2.3 3. RANSAC (20 points)

Rather than directly feeding all our keypoint matches into `fit_affine_matrix`, we can use **RANSAC** (“RANdom SAMple Consensus”) to select only “inliers” to use for computing the transformation matrix.

Use Euclidean distance as a measure of inliers vs. outliers.

The steps of RANSAC are: 1. Select random set of matches 2. Compute affine transformation matrix - You can call your `fit_affine_matrix` function for this. **Make sure to explicitly pass in `to_pad=False`**. 3. Find inliers using the given threshold 4. Repeat and keep the largest set of inliers (use `>`, i.e. break ties by whichever set is seen first) 5. Re-compute least-squares estimate on all of the inliers

Implement `ransac` below.

```
[13]: def ransac(keypoints1, keypoints2, matches, n_iters=200, threshold=20):
      """
      Use RANSAC to find a robust affine transformation:
```

1. Select random set of matches
2. Compute affine transformation matrix
3. Compute inliers via Euclidean distance
4. Keep the largest set of inliers (use `>`, i.e. break ties by whichever set is seen first)
5. Re-compute least-squares estimate on all of the inliers

Update `max_inliers` as a boolean array where `True` represents the keypoint at this index is an inlier, while `False` represents that it is not an inlier.

Hint:

You can use `fit_affine_matrix` to compute the affine transformation matrix.

Make sure to pass in `to_pad=False`, since we pad the matches for you here.

You can compute elementwise boolean operations between two numpy arrays, and use boolean arrays to select array elements by index:

<https://numpy.org/doc/stable/reference/arrays.indexing.html#boolean-array-indexing>

Args:

`keypoints1`: $M1 \times 2$ matrix, each row is a point

`keypoints2`: $M2 \times 2$ matrix, each row is a point

`matches`: $N \times 2$ matrix, each row represents a match
[index of keypoint1, index of keypoint 2]

`n_iters`: the number of iterations RANSAC will run

`threshold`: the threshold to find inliers

Returns:

`H`: a robust estimation of affine transformation from `keypoints2` to `keypoints 1`

"""

Copy matches array, to avoid overwriting it

`orig_matches = matches.copy()`

`matches = matches.copy()`

`N = matches.shape[0]`

`n_samples = int(N * 0.2)`

`matched1 = pad(keypoints1[matches[:,0]])`

`matched2 = pad(keypoints2[matches[:,1]])`

`max_inliers = np.zeros(N, dtype=bool)`

`n_inliers = 0`

```

# RANSAC iteration start

# Note: while there're many ways to do random sampling, we use
# `np.random.shuffle()` followed by slicing out the first `n_samples`
# matches here in order to align with the autograder.
# Sample with this code:
for i in range(n_iters):
    # 1. Select random set of matches
    np.random.shuffle(matches)
    samples = matches[:n_samples]
    sample1 = pad(keypoints1[samples[:,0]])
    sample2 = pad(keypoints2[samples[:,1]])

    ### YOUR CODE HERE

    # 2. Compute affine transformation matrix, map sample2 to sample1
    H = fit_affine_matrix(sample1, sample2, to_pad=False)

    # 3. Compute inliers via Euclidean distance
    dist = np.linalg.norm(matched1 @ H - matched2, axis=1)

    # 4. Keep the largest set of inliers
    max_inliers = max_inliers | (dist < threshold)

    ### END YOUR CODE

    # 5. Re-compute least-squares estimate on all of the inliers
    if max_inliers.any():
        H, _ = fit_affine_matrix(matched1[max_inliers], matched2[max_inliers],
        ↪to_pad=False), None
    else:
        # Fallback: use identity if no inliers found
        H = np.eye(3)

    return H, orig_matches[max_inliers]

```

Now, run through the following cells to get a panorama. You'll be able to see the difference from the result we got before without RANSAC.

```

[14]: # Set seed to compare output against solution image
np.random.seed(131)

H, robust_matches = ransac(keypoints1, keypoints2, matches, threshold=1)
print("Robust matches shape = ", robust_matches.shape)

```

```

print("H = \n", H)

# Visualize robust matches
fig, ax = plt.subplots(1, 1, figsize=(12, 9))
plot_matches(ax, img1, img2, keypoints1, keypoints2, robust_matches)
plt.axis('off')
plt.title('RANSAC Robust Matches')
plt.show()

plt.imshow(imread('solution_ransac.png'))
plt.axis('off')
plt.title('RANSAC Robust Matches Solution')
plt.show()

```

Robust matches shape = (0, 2)

H =

```

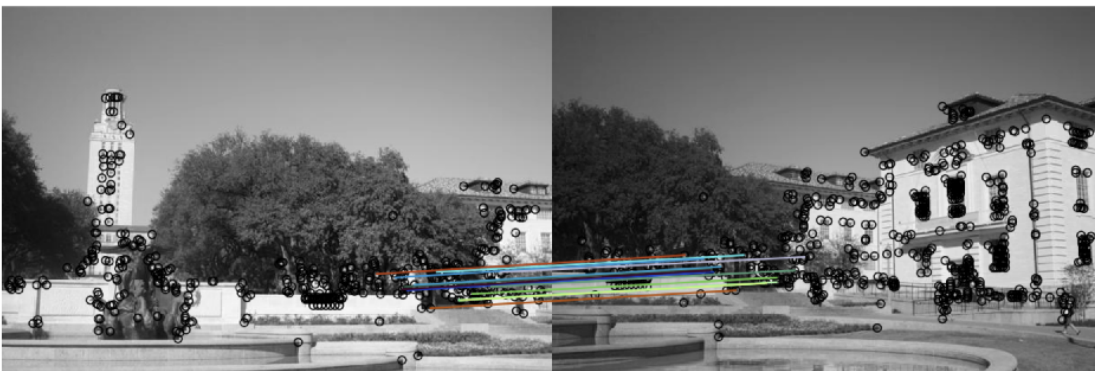
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]

```

RANSAC Robust Matches



RANSAC Robust Matches Solution



We can now use the transformation matrix H computed using the robust matches to warp our images and create a better-looking panorama.

```
[15]: output_shape, offset = get_output_space(img1, [img2], [H])

# Warp images into output space
img1_warped = warp_image(img1, np.eye(3), output_shape, offset)
img1_mask = (img1_warped != -1) # Mask == 1 inside the image
img1_warped[~img1_mask] = 0      # Return background values to 0

img2_warped = warp_image(img2, H, output_shape, offset)
img2_mask = (img2_warped != -1) # Mask == 1 inside the image
img2_warped[~img2_mask] = 0      # Return background values to 0

# Plot warped images
plt.subplot(1,2,1)
plt.imshow(img1_warped)
plt.title('Image 1 warped')
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(img2_warped)
plt.title('Image 2 warped')
plt.axis('off')

plt.show()
```

Image 1 warped



Image 2 warped



```
[16]: # Merge the two images
merged = img1_warped + img2_warped

# Track the overlap by adding the masks together
overlap = (img1_mask * 1.0 + # Multiply by 1.0 for bool -> float conversion
```

```

img2_mask)

# Normalize through division by `overlap` - but ensure the minimum is 1
normalized = merged / np.maximum(overlap, 1)
plt.imshow(normalized)
plt.axis('off')
plt.title('RANSAC Robust Panorama')
plt.show()

plt.imshow(imread('solution_ransac_panorama.png'))
plt.axis('off')
plt.title('RANSAC Robust Panorama Solution')
plt.show()

```

RANSAC Robust Panorama





2.4 4. Better Image Merging (10 points)

You will notice the blurry region and unpleasant lines in the middle of the final panoramic image. Using a very simple technique called linear blending, we can smooth out a lot of these artifacts.

Currently, all the pixels in the overlapping region are weighted equally. However, since the pixels at the left and right ends of the overlap are very well complemented by the pixels in the other image, they can be made to contribute less to the final panorama.

Linear blending can be done with the following steps:

1. Define left and right margins for blending to occur between
2. Define a weight matrix for image 1 such that:
 - From the left of the output space to the left margin the weight is 1
 - From the left margin to the right margin, the weight linearly decrements from 1 to 0
3. Define a weight matrix for image 2 such that:
 - From the right of the output space to the right margin the weight is 1
 - From the left margin to the right margin, the weight linearly increments from 0 to 1
4. Apply the weight matrices to their corresponding images
5. Combine the images

In `linear_blend` below, implement the linear blending scheme to make the panorama look more natural.

```
[25]: def linear_blend(img1_warped, img2_warped):
      """
      Linearly blend img1_warped and img2_warped by following the steps:

      1. Define left and right margins (already done for you)
      2. Define a weight matrices for img1_warped and img2_warped
```

- np.linspace and np.tile functions will be useful*
3. Apply the weight matrices to their corresponding images
 4. Combine the images

Args:

img1_warped: Reference image warped into output space
img2_warped: Transformed image warped into output space

Returns:

merged: Merged image in output space

```

"""
out_H, out_W = img1_warped.shape # Height and width of output space
img1_mask = (img1_warped != 0) # Mask == 1 inside the image
img2_mask = (img2_warped != 0) # Mask == 1 inside the image

# Find column of middle row where warped image 1 ends
# This is where to end weight mask for warped image 1
right_margin = out_W - np.argmax(np.fliplr(img1_mask)[out_H//2, :]).
↳reshape(1, out_W), 1)[0]

# Find column of middle row where warped image 2 starts
# This is where to start weight mask for warped image 2
left_margin = np.argmax(img2_mask[out_H//2, :].reshape(1, out_W), 1)[0]

# Initialize weight maps
w1 = np.zeros((out_H, out_W), dtype=np.float32)
w2 = np.zeros((out_H, out_W), dtype=np.float32)

### YOUR CODE HERE
w1[:, :left_margin] = 1
w1[:, left_margin:right_margin] = np.linspace(1, 0, right_margin -
↳left_margin)
w2[:, right_margin:] = 1
w2[:, left_margin:right_margin] = np.linspace(0, 1, right_margin -
↳left_margin)
img1_warped = img1_warped * w1
img2_warped = img2_warped * w2
merged = img1_warped + img2_warped

### END YOUR CODE

return merged

```

Now let's see how linear blending improves our result.

```

[26]: img1 = imread('uttower1.jpg', as_gray=True)
img2 = imread('uttower2.jpg', as_gray=True)

# Set seed to compare output against solution
np.random.seed(131)

# Detect keypoints in both images
ec1_keypoints1 = corner_peaks(harris_corners(img1, window_size=3),
                              threshold_rel=0.05,
                              exclude_border=8)
ec1_keypoints2 = corner_peaks(harris_corners(img2, window_size=3),
                              threshold_rel=0.05,
                              exclude_border=8)

print("EC1 keypoints1 shape = ", ec1_keypoints1.shape)
print("EC1 keypoints2 shape = ", ec1_keypoints2.shape)

# Extract features from the corners
ec1_desc1 = describe_keypoints(img1, ec1_keypoints1,
                                desc_func=simple_descriptor,
                                patch_size=16)
ec1_desc2 = describe_keypoints(img2, ec1_keypoints2,
                                desc_func=simple_descriptor,
                                patch_size=16)

print("EC1 desc1 shape = ", ec1_desc1.shape)
print("EC1 desc2 shape = ", ec1_desc2.shape)

# Match descriptors in image1 to those in image2
ec1_matches = match_descriptors(ec1_desc1, ec1_desc2, 0.7)

H, robust_matches = ransac(ec1_keypoints1, ec1_keypoints2, ec1_matches,
    ↪ threshold=1)
print("Robust matches shape = ", robust_matches.shape)
print("H = \n", H)

output_shape, offset = get_output_space(img1, [img2], [H])
print("Output shape:", output_shape)
print("Offset:", offset)

# Warp images into output space
img1_warped = warp_image(img1, np.eye(3), output_shape, offset)
img1_mask = (img1_warped != -1) # Mask == 1 inside the image
img1_warped[~img1_mask] = 0     # Return background values to 0

img2_warped = warp_image(img2, H, output_shape, offset)
img2_mask = (img2_warped != -1) # Mask == 1 inside the image

```

```

img2_warped[~img2_mask] = 0      # Return background values to 0

# Merge the warped images using linear blending scheme
merged = linear_blend(img1_warped, img2_warped)

plt.imshow(merged)
plt.axis('off')
plt.title('Linear Blend')
plt.show()

plt.imshow(imread('solution_linear_blend.png'))
plt.axis('off')
plt.title('Linear Blend Solution')
plt.show()

```

```

EC1 keypoints1 shape = (396, 2)
EC1 keypoints2 shape = (627, 2)
EC1 desc1 shape = (396, 256)
EC1 desc2 shape = (627, 256)
Robust matches shape = (0, 2)
H =
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
Output shape: [410 615]
Offset: [0. 0.]

```

Linear Blend



Linear Blend Solution



```
[27]: !apt-get update -qq  
      !apt-get install -y texlive-xetex texlive-fonts-recommended_  
      ↪texlive-plain-generic pandoc
```

```
!apt-get install -y pandoc
!pip install nbconvert
```

W: Skipping acquire of configured file 'main/source/Sources' as repository 'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide it (sources.list entry misspelt?)

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tiutils
teckit tex-common tex-gyre texlive-base texlive-binaries texlive-latex-base
texlive-latex-extra texlive-latex-recommended texlive-pictures tipa
xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tiutils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 53 newly installed, 0 to remove and 46 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1 [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all 0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17 [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all 20200910-1 [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1 [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64 0.35-15build2 [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64 0.19-3build2 [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]

Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64 1.0.2-1build4 [45.2 kB]

Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64 2.13.1-1 [1,221 kB]

Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all 2.004.5-6.1 [4,532 kB]

Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all 20201225-1build1 [397 kB]

Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all 20180621-3.1 [10.2 MB]

Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java all 18-1 [4,720 B]

Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-java all 43-1 [10.8 kB]

Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-java all 1.2-2 [60.3 kB]

Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]

Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration all 1.18 [5,336 B]

Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64 3.0.2-7ubuntu2.11 [50.1 kB]

Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems all 3.3.5-2ubuntu1.2 [228 kB]

Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
0.1.1-2 [12.6 kB]
Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmllrpc all
0.3.2-1ubuntu0.1 [24.9 kB]
Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
amd64 3.0.2-7ubuntu2.11 [5,114 kB]
Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsyntax2
amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
2.5.11+ds1-1 [421 kB]
Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2
amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64
0.13.72+dfsg.1-1.1 [27.0 kB]
Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:37 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:38 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:39 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:40 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 <http://archive.ubuntu.com/ubuntu> jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:47 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]

```

Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 9s (20.3 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 125080 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.

```

```

Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...

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Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluaajit2:amd64.
Preparing to unpack
.../32-libtexluaajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...

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Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../45-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../46-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../47-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../48-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../49-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../50-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../51-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../52-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...

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Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-

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ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

```

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

Processing triggers for tex-common (6.17) ...

Running updmf-sys. This may take some time... done.

Running mktexlsr /var/lib/texmf ... done.

Building format(s) --all.

This may take some time... done.

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

0 upgraded, 0 newly installed, 0 to remove and 46 not upgraded.

Requirement already satisfied: nbconvert in /usr/local/lib/python3.12/dist-packages (7.16.6)

Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (4.13.5)

Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->nbconvert) (6.3.0)

Requirement already satisfied: defusedxml in /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.7.1)

Requirement already satisfied: jinja2>=3.0 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.1.6)

Requirement already satisfied: jupyter-core>=4.7 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (5.9.1)

Requirement already satisfied: jupyterlab-pygments in /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.3.0)

Requirement already satisfied: markupsafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.0.3)

Requirement already satisfied: mistune<4,>=2.0.3 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.1.4)

Requirement already satisfied: nbclient>=0.5.0 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.10.2)

Requirement already satisfied: nbformat>=5.7 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (5.10.4)

Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from nbconvert) (25.0)

Requirement already satisfied: pandocfilters>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (1.5.1)

Requirement already satisfied: pygments>=2.4.1 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (2.19.2)

Requirement already satisfied: traitlets>=5.1 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (5.7.1)

Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert) (0.5.1)

Requirement already satisfied: tinycss2<1.5,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->nbconvert) (1.4.0)

Requirement already satisfied: platformdirs>=2.5 in /usr/local/lib/python3.12/dist-packages (from jupyter-core>=4.7->nbconvert) (4.5.0)

Requirement already satisfied: jupyter-client>=6.1.12 in /usr/local/lib/python3.12/dist-packages (from nbclient>=0.5.0->nbconvert) (7.4.9)

Requirement already satisfied: fastjsonschema>=2.15 in /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert) (2.21.2)

Requirement already satisfied: jsonschema>=2.6 in /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert) (4.25.1)

Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->nbconvert) (2.8)

Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->nbconvert) (4.15.0)

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (25.4.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (2025.9.1)

Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.37.0)

Requirement already satisfied: rpds-py>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.28.0)

Requirement already satisfied: entrypoints in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (0.4)

Requirement already satisfied: nest-asyncio>=1.5.4 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (1.6.0)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (2.9.0.post0)

Requirement already satisfied: pyzmq>=23.0 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (26.2.1)

Requirement already satisfied: tornado>=6.2 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (6.5.1)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (1.17.0)

```
[ ]: !jupyter nbconvert --to pdf "HW4.ipynb"
```